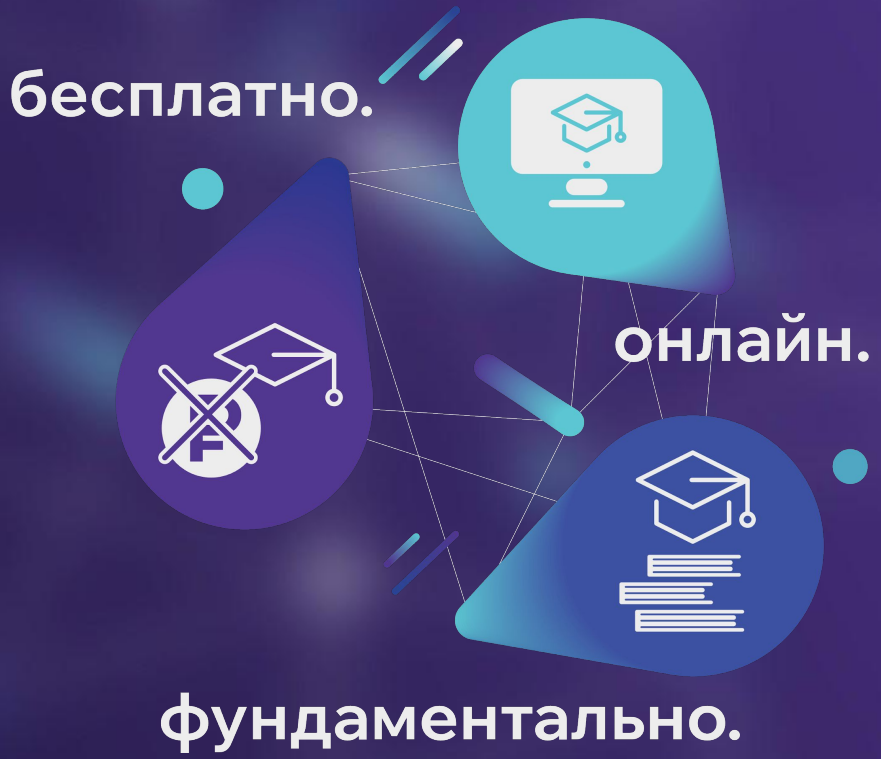
Deep Learning School



DLS



Diffusion models

Коновалова Нина

@ReadingAI

@AfeliaN

DLS

PART 1

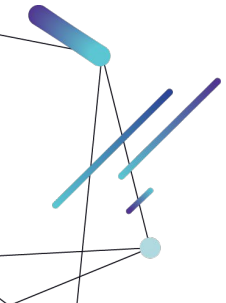
Diffusion models

Basics: forward & reverse processes, training, guidance and text-to-image



Agenda

01	**Forward process**	How noise is added: variance exploding vs variance preserving
02	**Reverse process**	Step-by-step denoising and how it compares to a VAE
03	**Training**	ELBO, the simplified loss and the training / sampling algorithms
04	**DDPM & guidance**	Samples in pixel space, classifier and classifier-free guidance
05	**Text-to-image**	GLIDE, Imagen, Dalle2



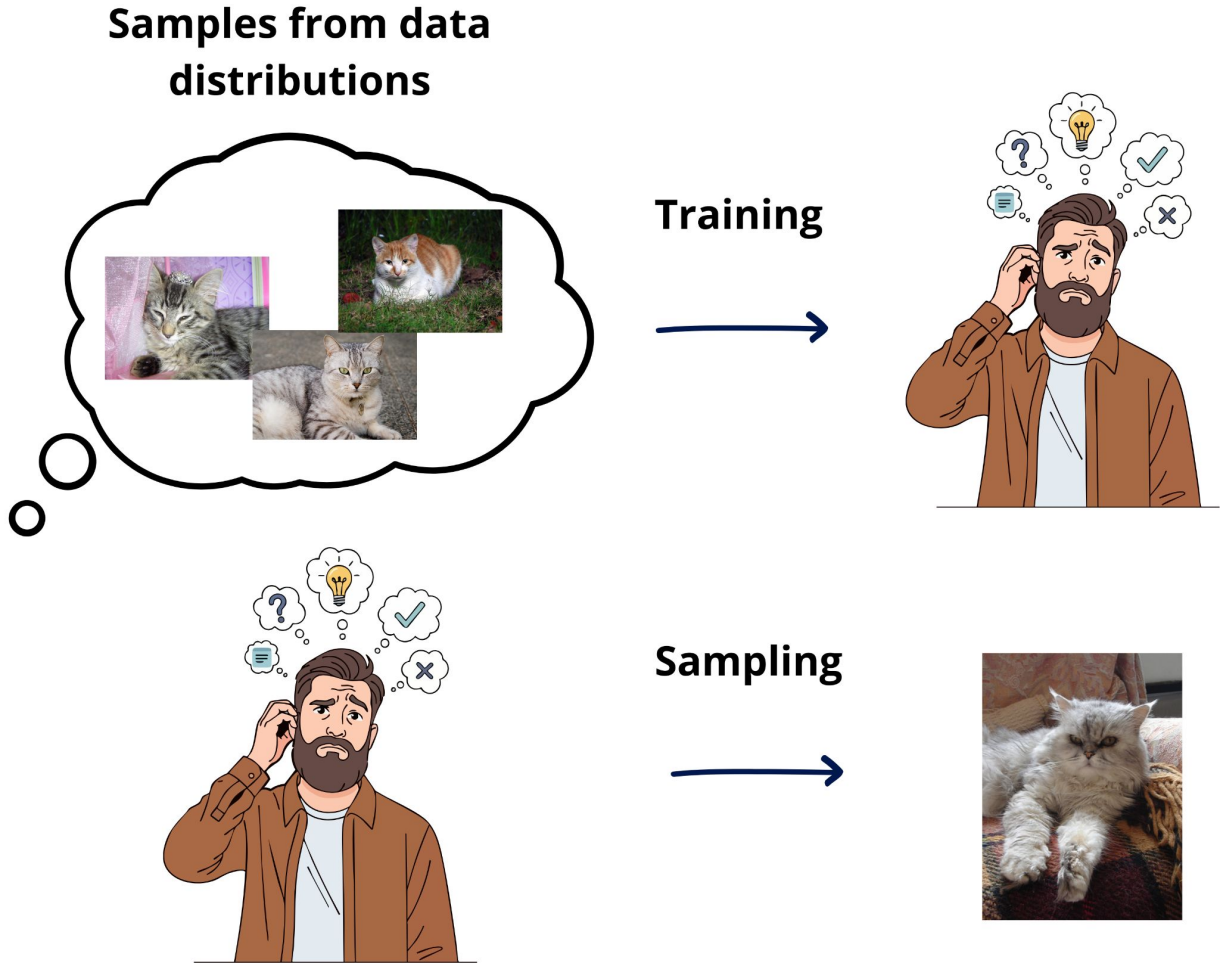
Generative models

1 Training

Learn the data distribution p(x) from a set of samples

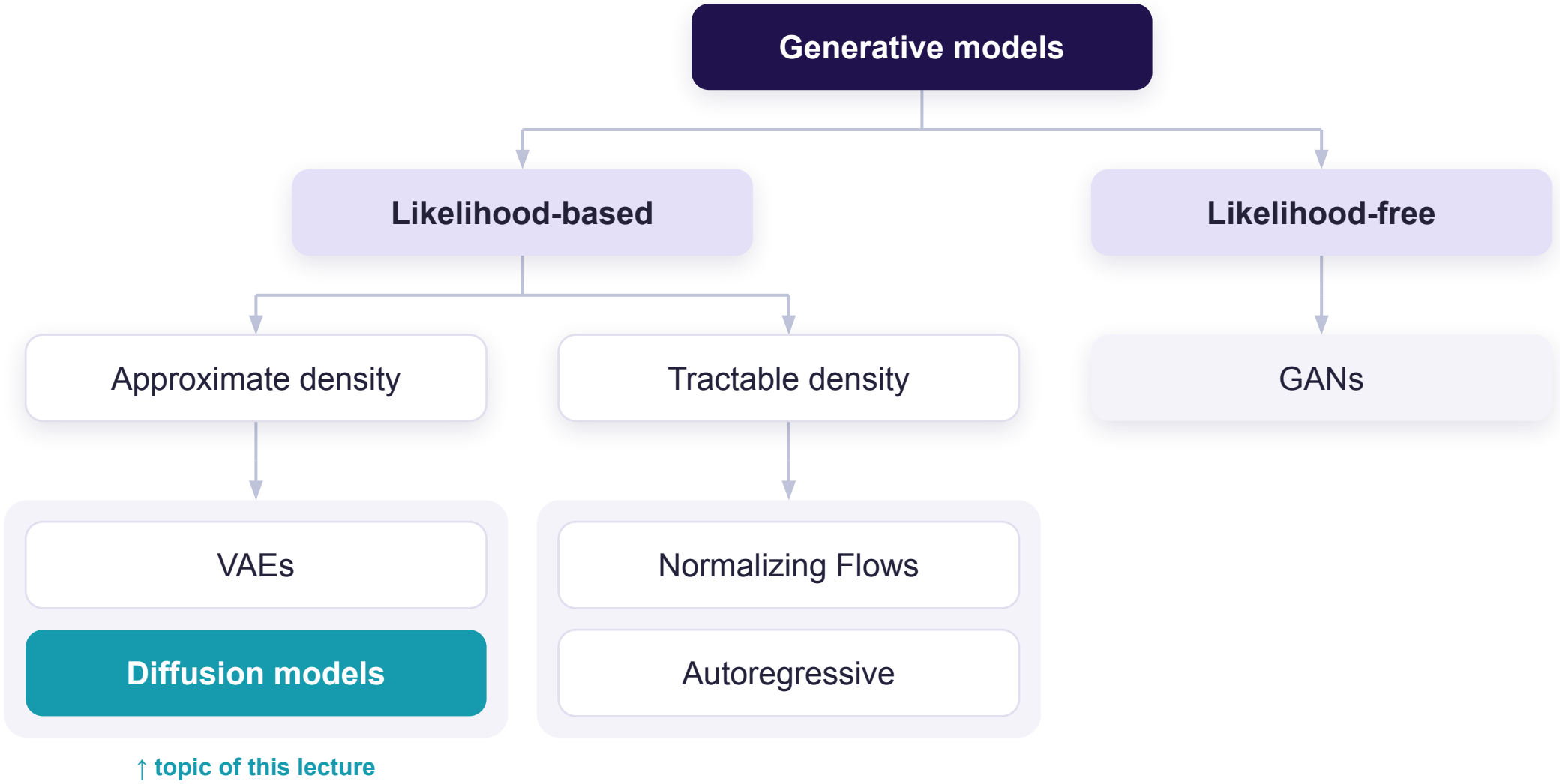
2 Sampling

Draw new, previously unseen samples from the learned distribution



Images: FLUX.2 [klein]

Generative models: a taxonomy



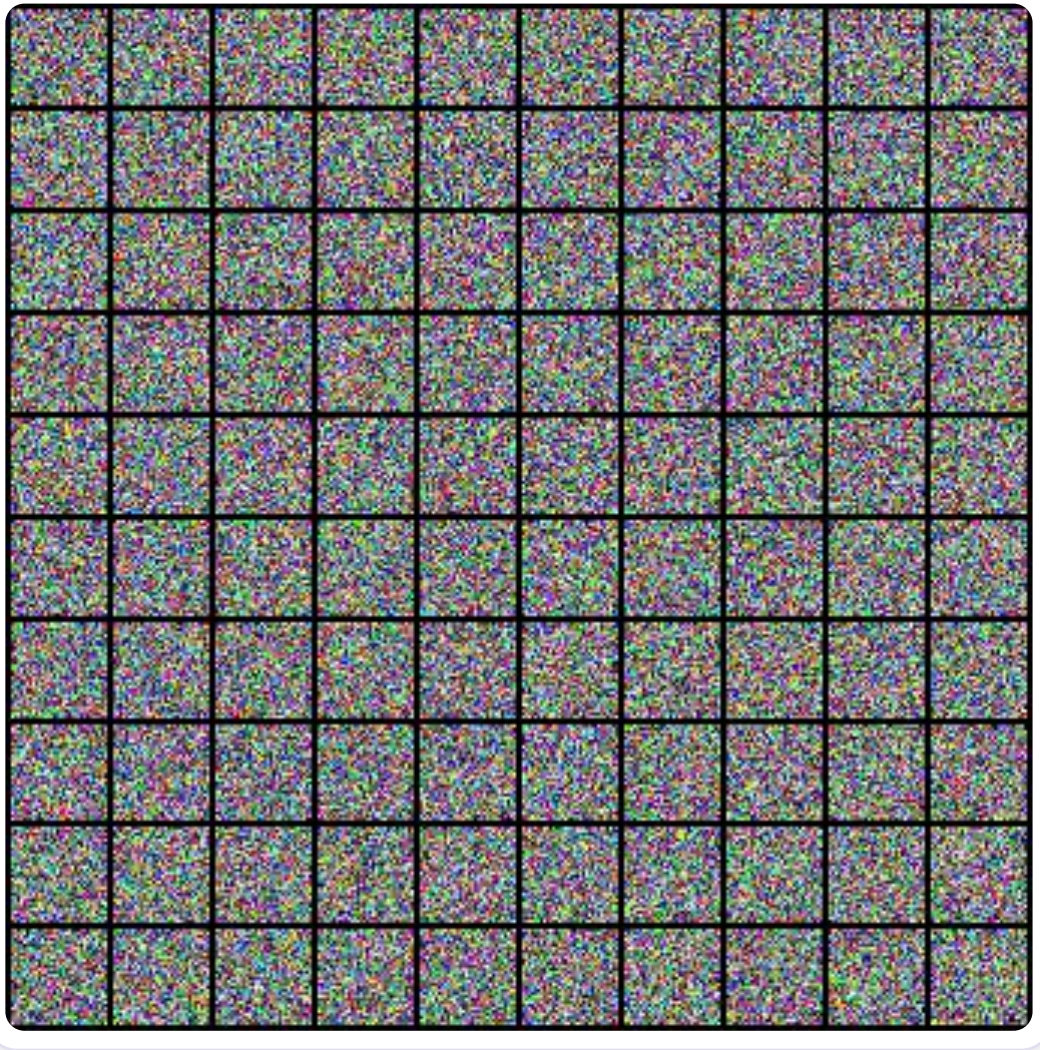
StyleGAN3 vs StyleGAN2

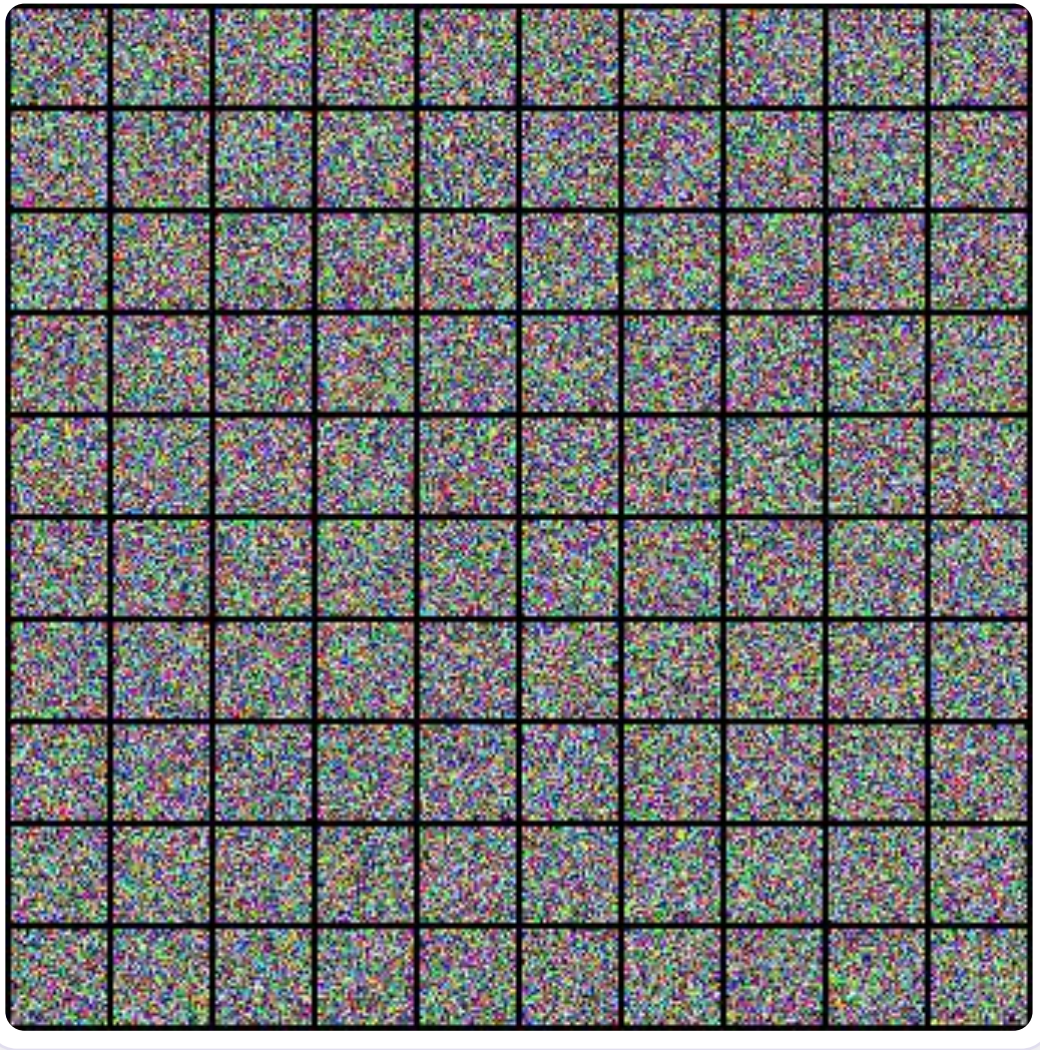




Source: nvlabs.github.io/stylegan3

Diffusion models





Diffusion models: the idea

→ Forward process

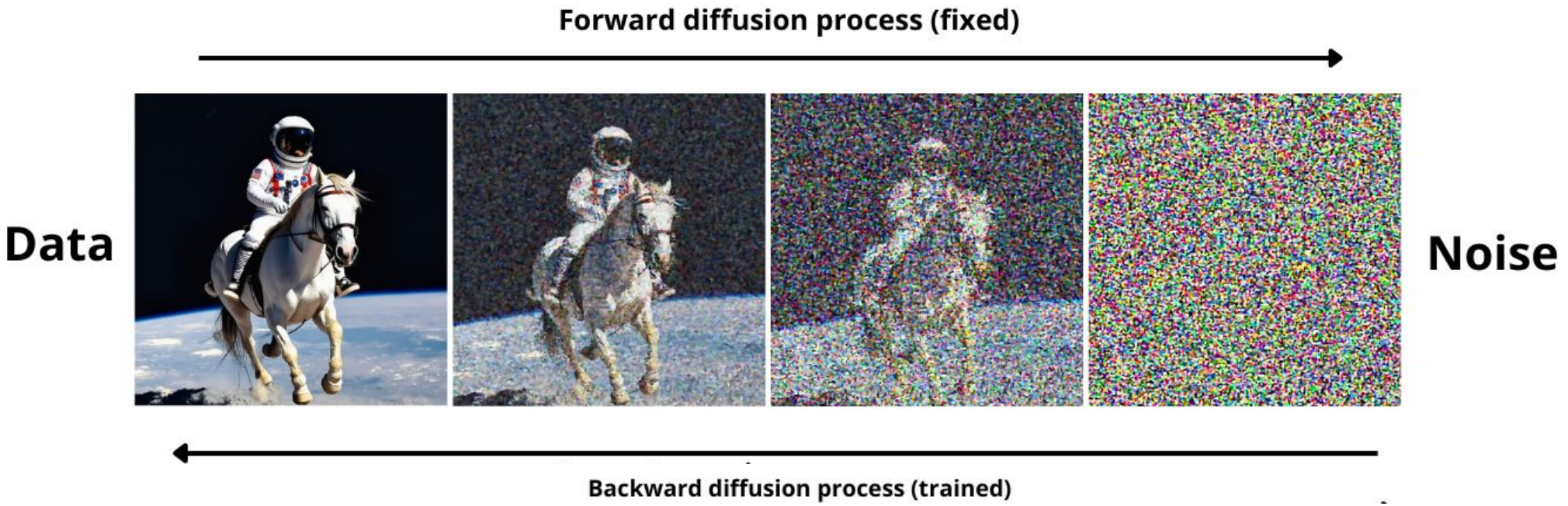
fixed

Gradually adds noise to the input

← Reverse process

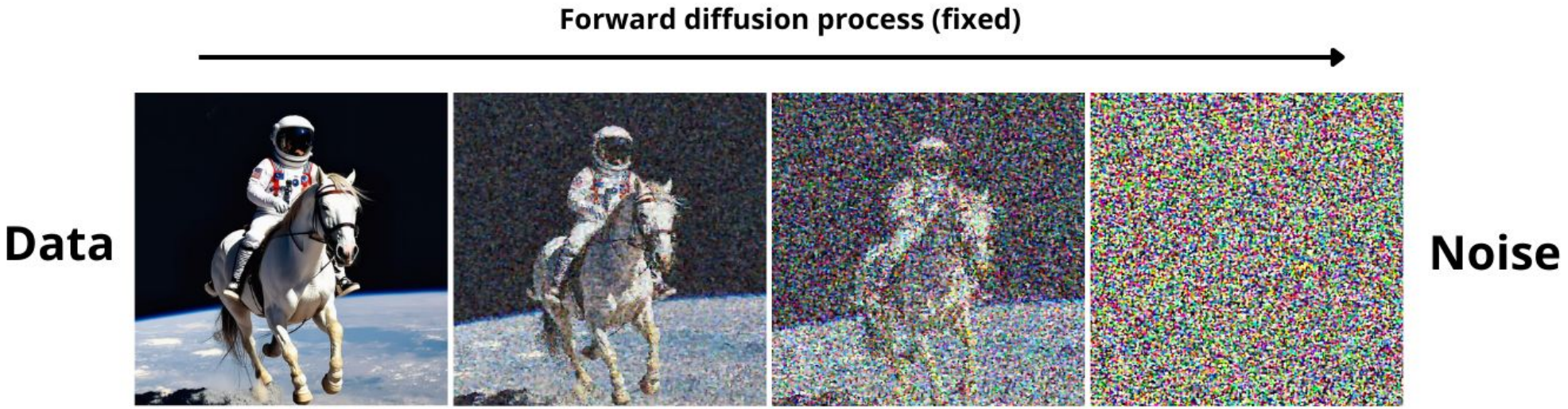
trained

Learns to generate data by denoising



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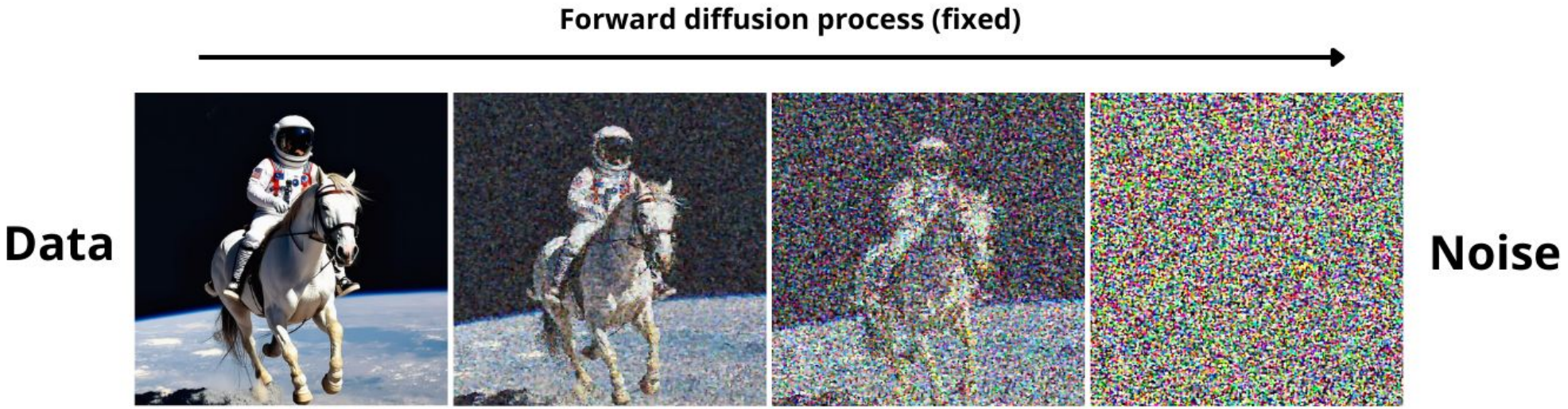
Forward process



Variance Exploding (VE)

Variance Preserving (VP)

Forward process



START

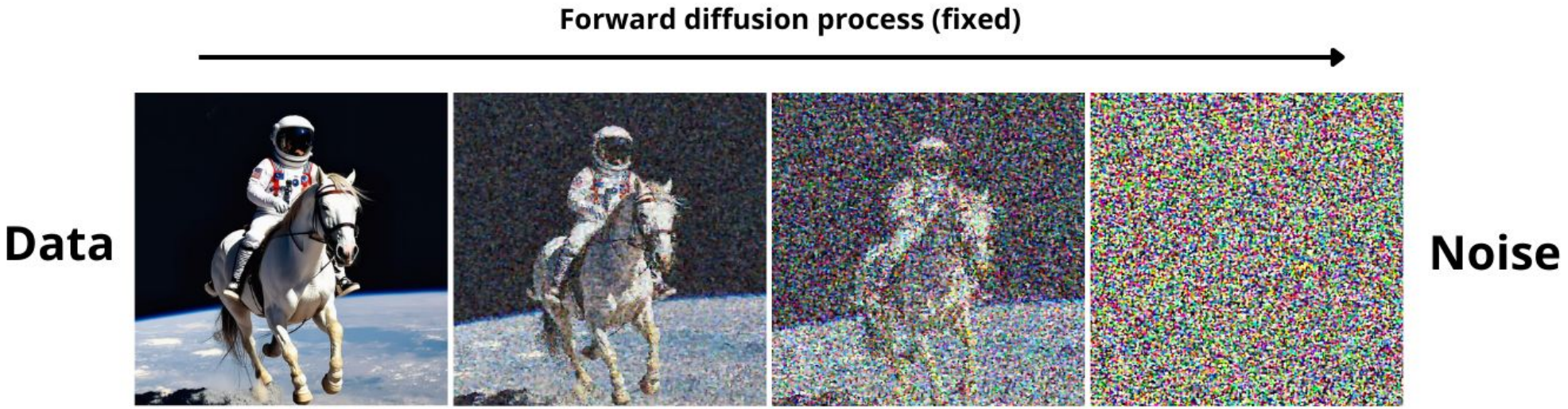
Variance Exploding (VE)

$$x \sim p_{data}$$

Variance Preserving (VP)

$$x \sim p_{data}$$

Forward process



SAMPLE

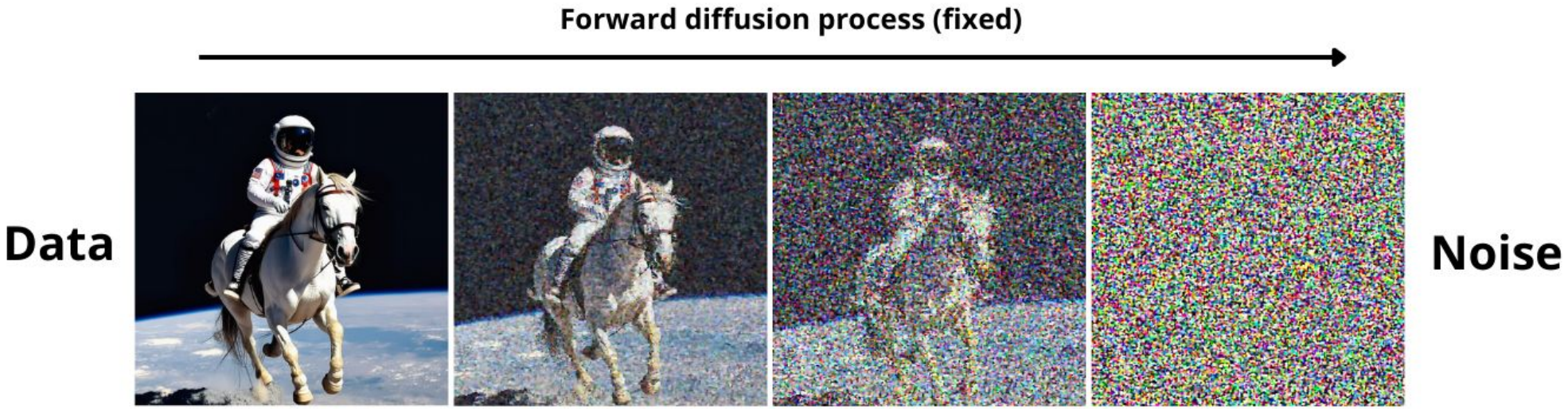
Variance Exploding (VE)

$$x_t = x_{t-1} + g(t)\epsilon_t$$
$$\epsilon_t \sim \mathcal{N}(0, I)$$

Variance Preserving (VP)

$$x_t = \sqrt{1-\beta_t}x_{t-1} + \sqrt{\beta_t}\epsilon_t$$
$$\epsilon_t \sim \mathcal{N}(0, I)$$

Forward process



SAMPLE

Variance Exploding (VE)

$$x_t = x_0 + \sigma_t \epsilon_t$$

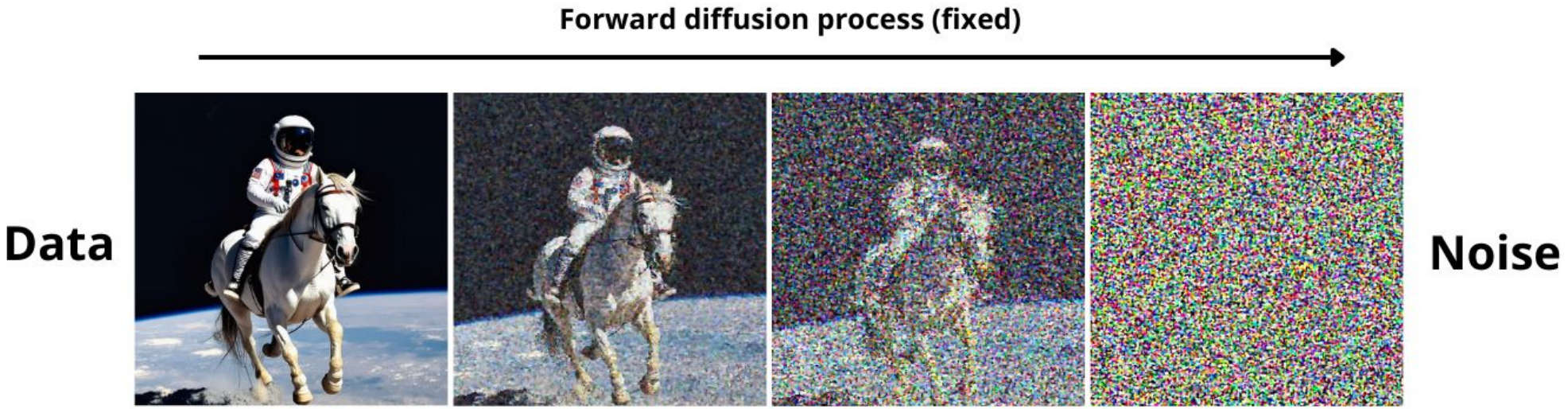
$$\sigma_t^2 = \sum_0^t g_i^2(i)$$

Variance Preserving (VP)

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon_t$$

$$\alpha_t = 1 - \beta_t \quad \bar{\alpha}_t = \prod_{i=1}^{t} \alpha_t$$

Forward process



DISTRIBUTION

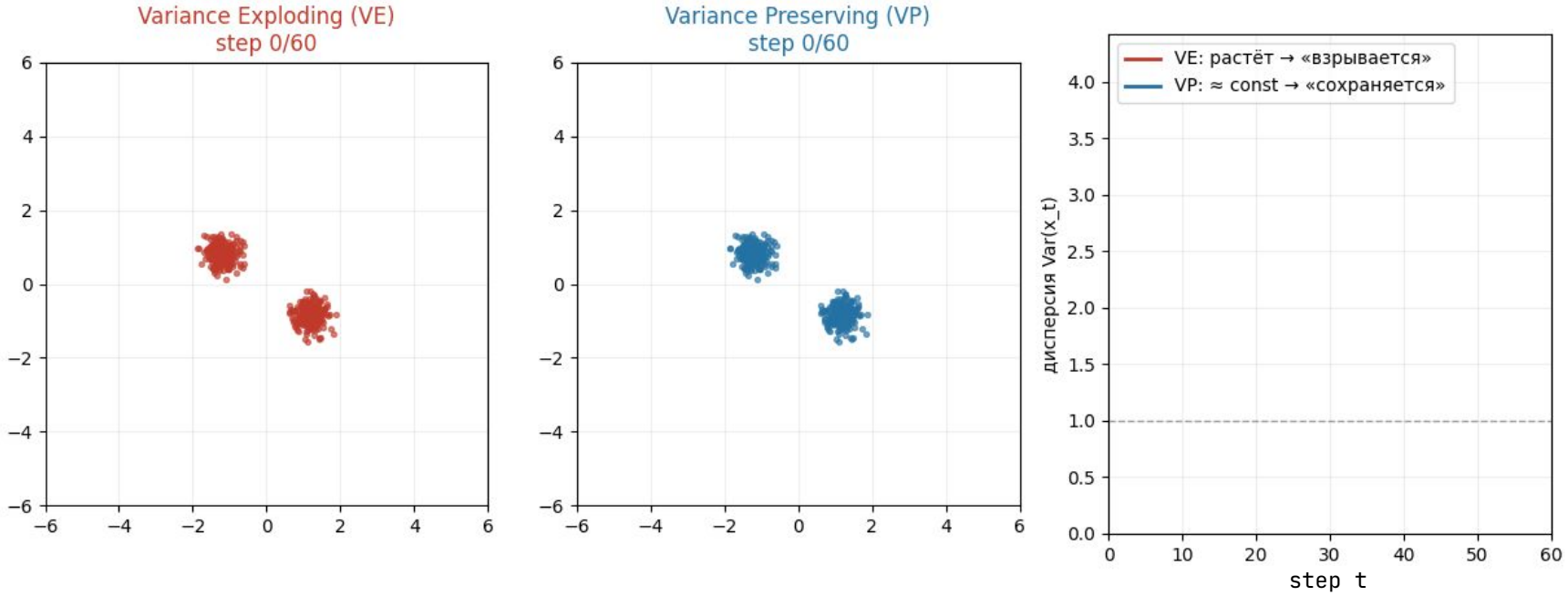
Variance Exploding (VE)

$$\mathcal{N}(x_t|x_0, \sigma_T^2 I)$$

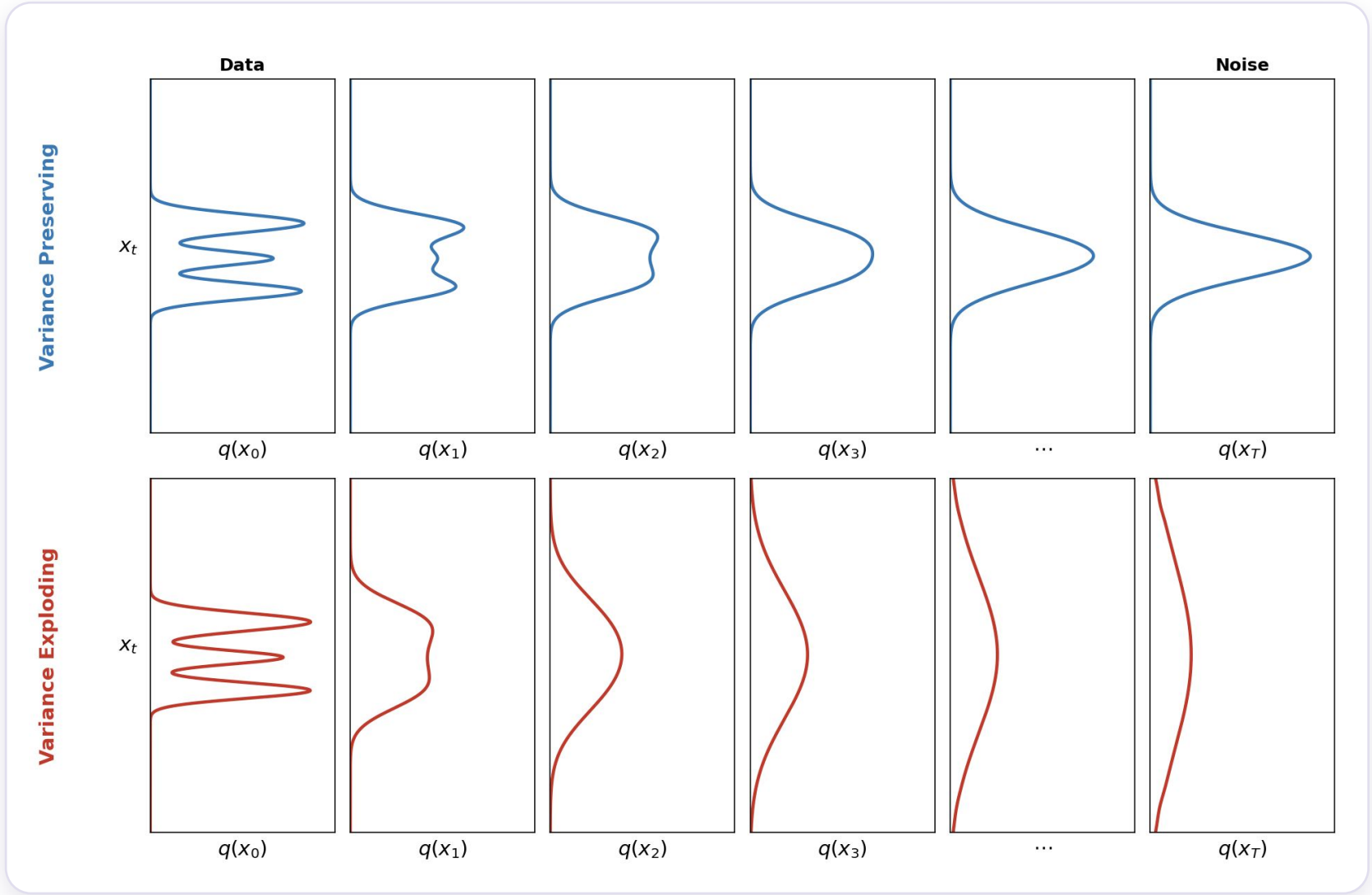
Variance Preserving (VP)

$$\mathcal{N}(x_t|\sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I)$$

Forward process: VE vs VP in action



Forward process: marginal distributions

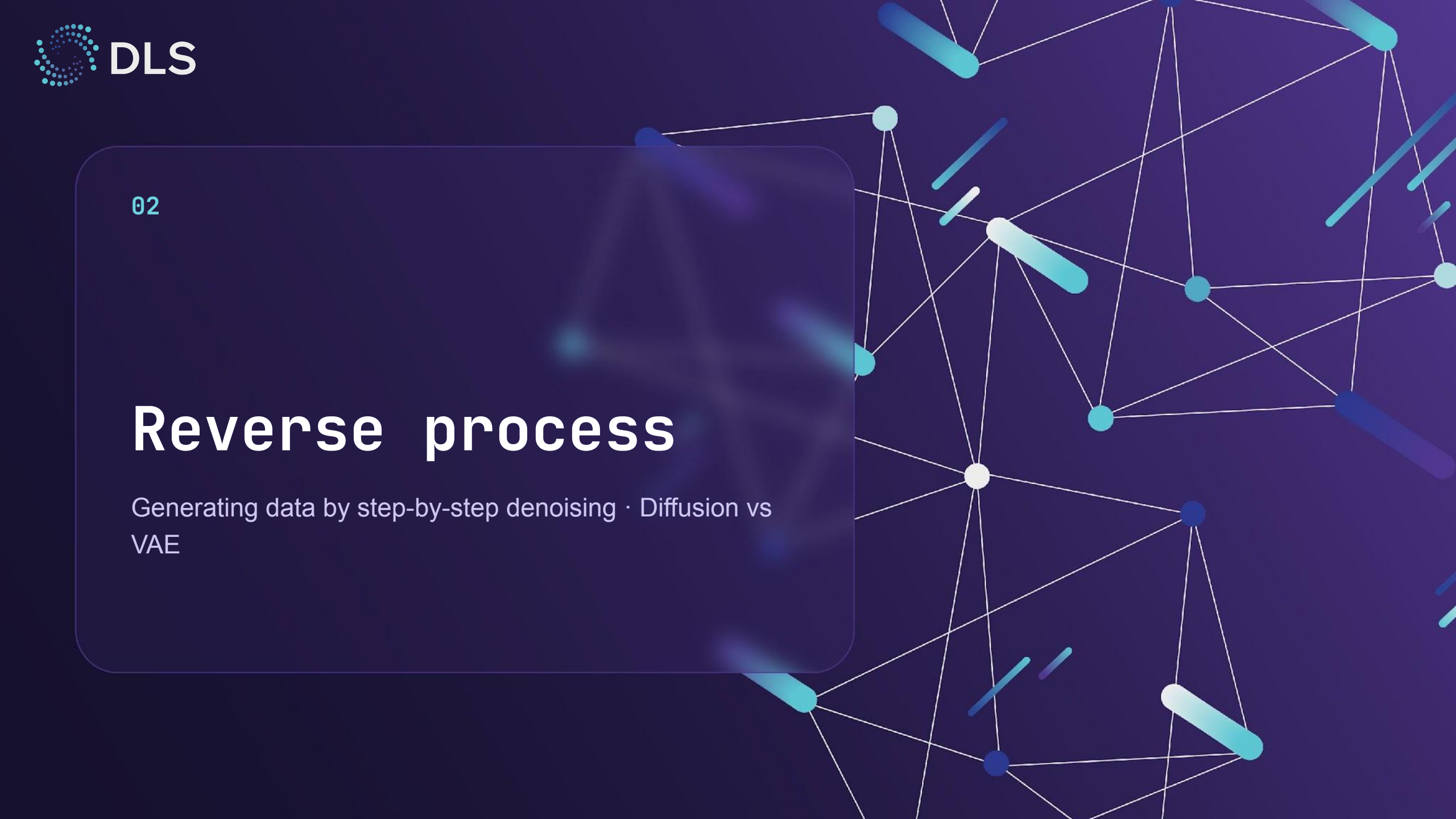


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02

Reverse process

Generating data by step-by-step denoising · Diffusion vs VAE



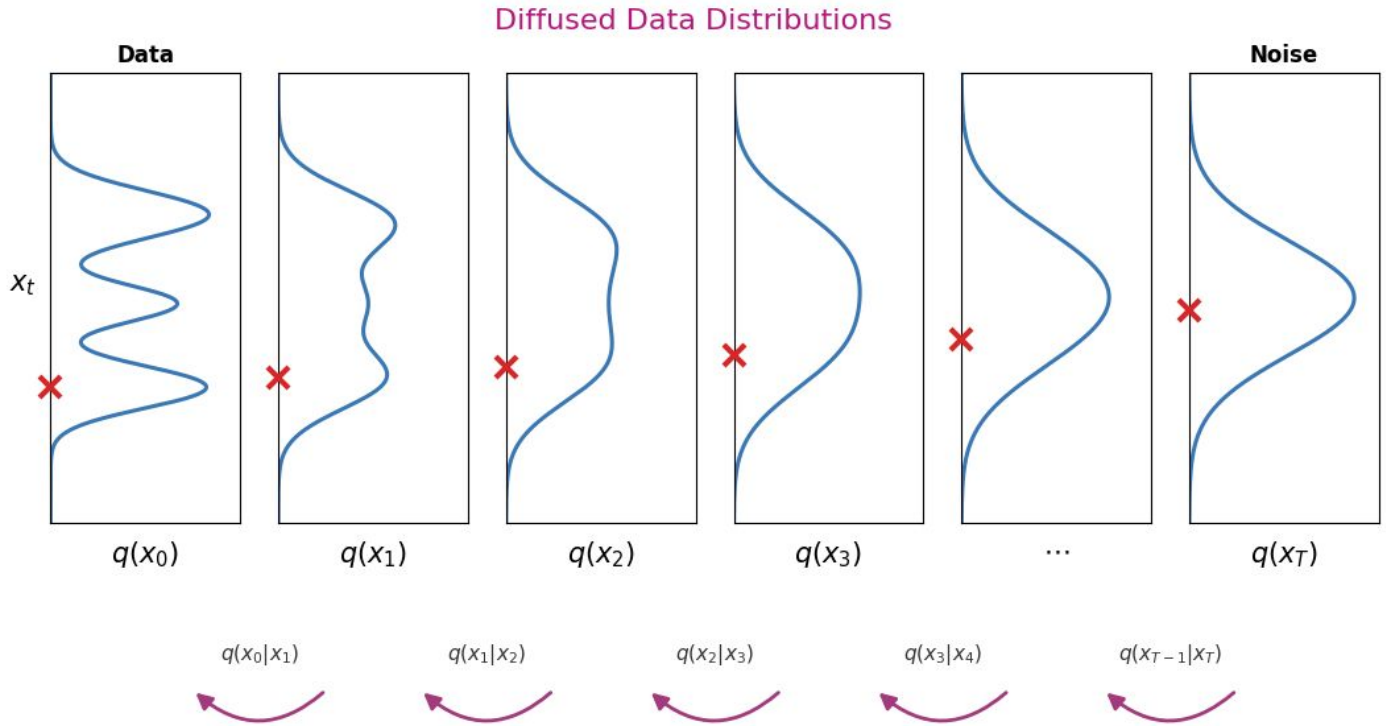
Generation by step-by-step denoising

1 **Sample pure noise**

$$\mathbf{x}_T \sim \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$$

2 **Iteratively denoise**

$$\mathbf{x}_{t-1} \sim \underbrace{q(\mathbf{x}_{t-1}|\mathbf{x}_t)}_{\text{True Denoising Dist.}}$$



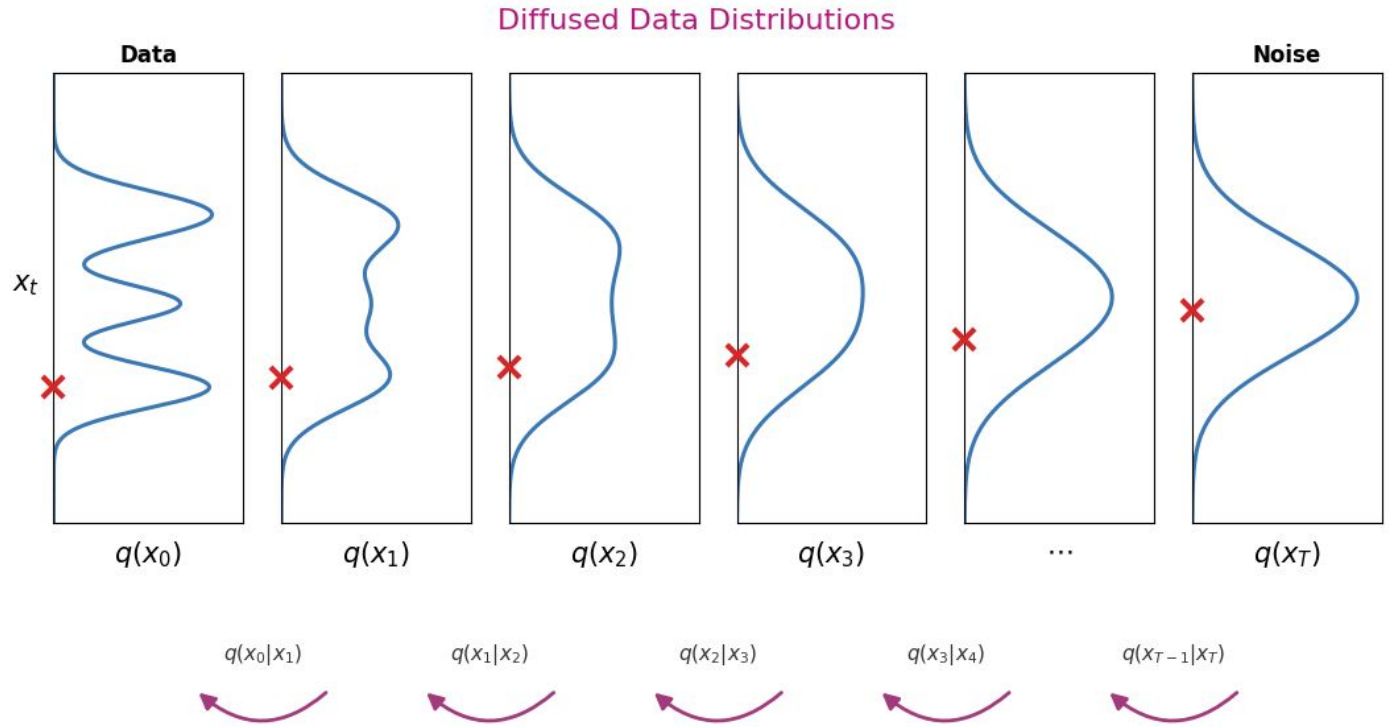
Generation by step-by-step denoising

1 **Sample pure noise**

$$\mathbf{x}_T \sim \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$$

2 **Iteratively denoise**

$$\mathbf{x}_{t-1} \sim \underbrace{q(\mathbf{x}_{t-1}|\mathbf{x}_t)}_{\text{True Denoising Dist.}}$$



$$q(x_{t-1} \mid x_t) = \int q(x_{t-1} \mid x_t, x_0) q(x_0 \mid x_t)\, dx_0 \longrightarrow \text{intractable :(}$$

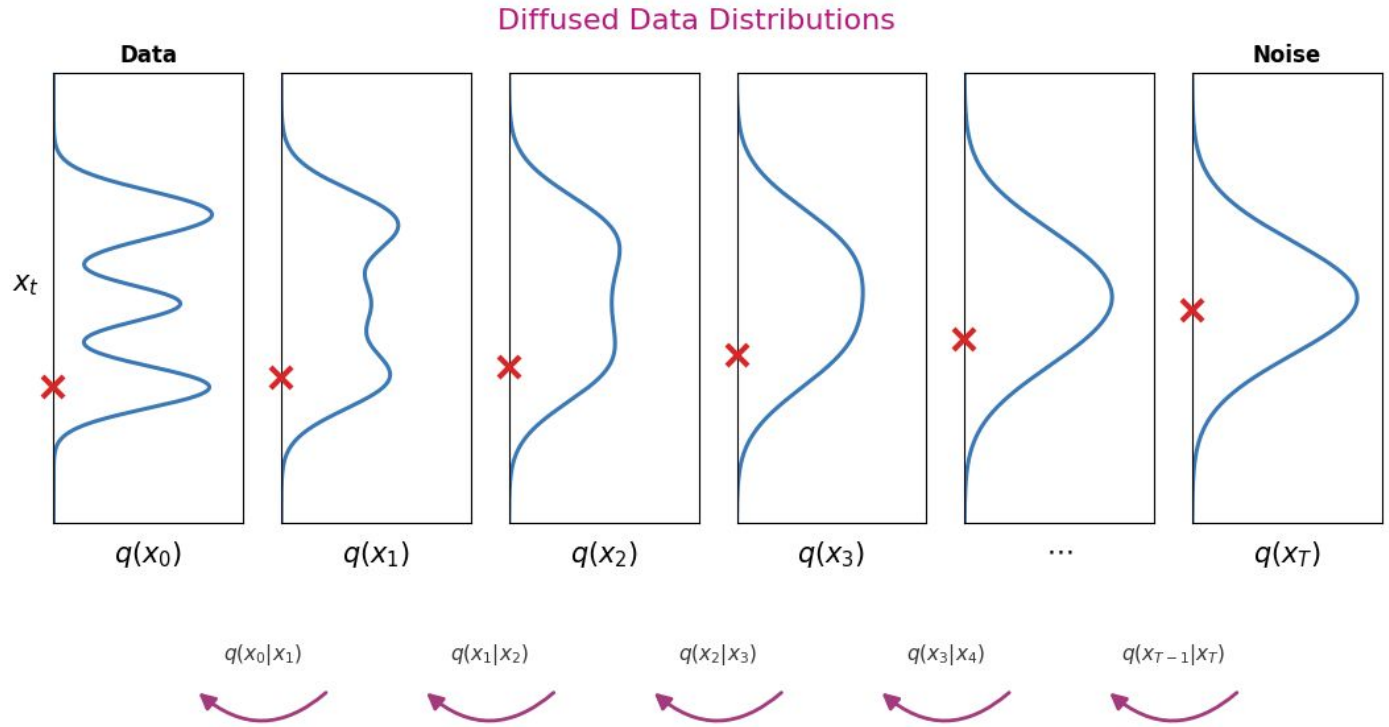
Generation by step-by-step denoising

1 **Sample pure noise**

$$\mathbf{x}_T \sim \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$$

2 **Iteratively denoise**

$$\mathbf{x}_{t-1} \sim \underbrace{q(\mathbf{x}_{t-1}|\mathbf{x}_t)}_{\text{True Denoising Dist.}}$$



$$q(x_{t-1} \mid x_t) = \int q(x_{t-1} \mid x_t, x_0) q(x_0 \mid x_t)\, dx_0 \longrightarrow$$ **intractable :(**

But can we approximate?

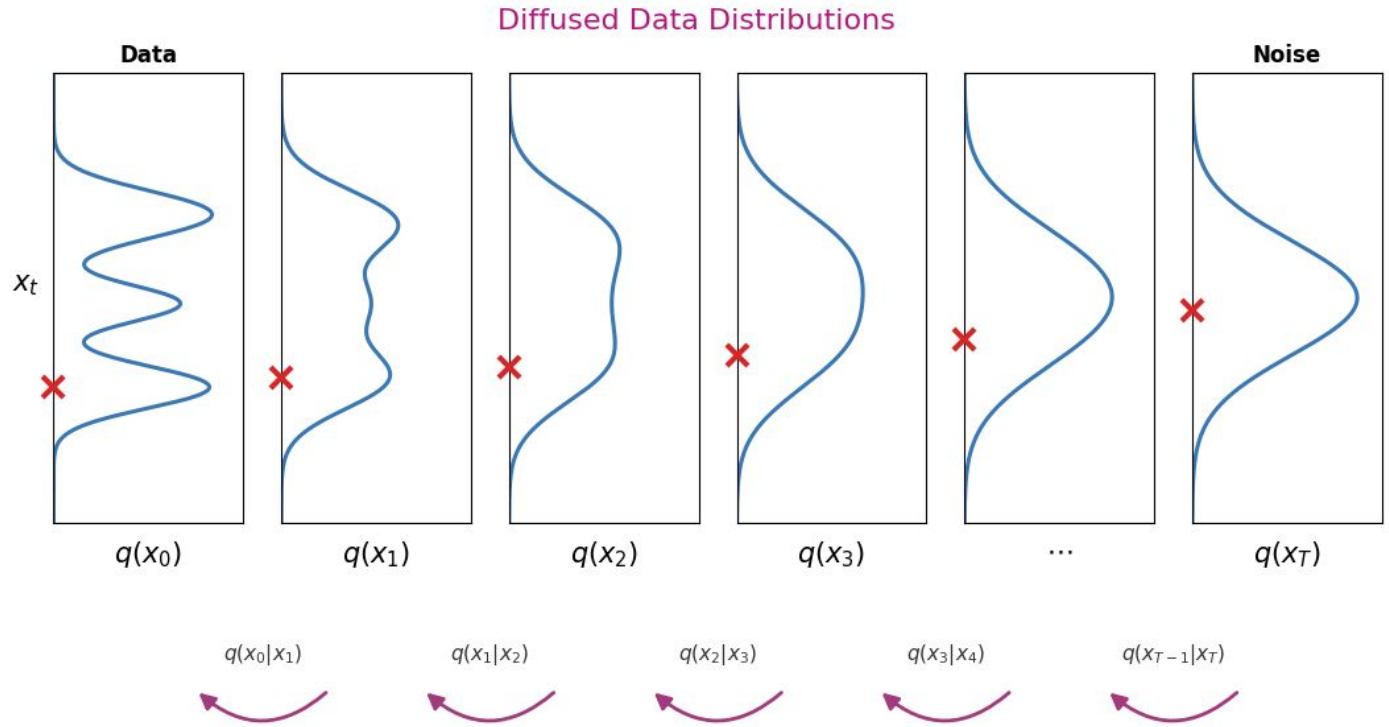
Generation by step-by-step denoising

1 **Sample pure noise**

$$\mathbf{x}_T \sim \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$$

2 **Iteratively denoise**

$$\mathbf{x}_{t-1} \sim \underbrace{q(\mathbf{x}_{t-1}|\mathbf{x}_t)}_{\text{True Denoising Dist.}}$$

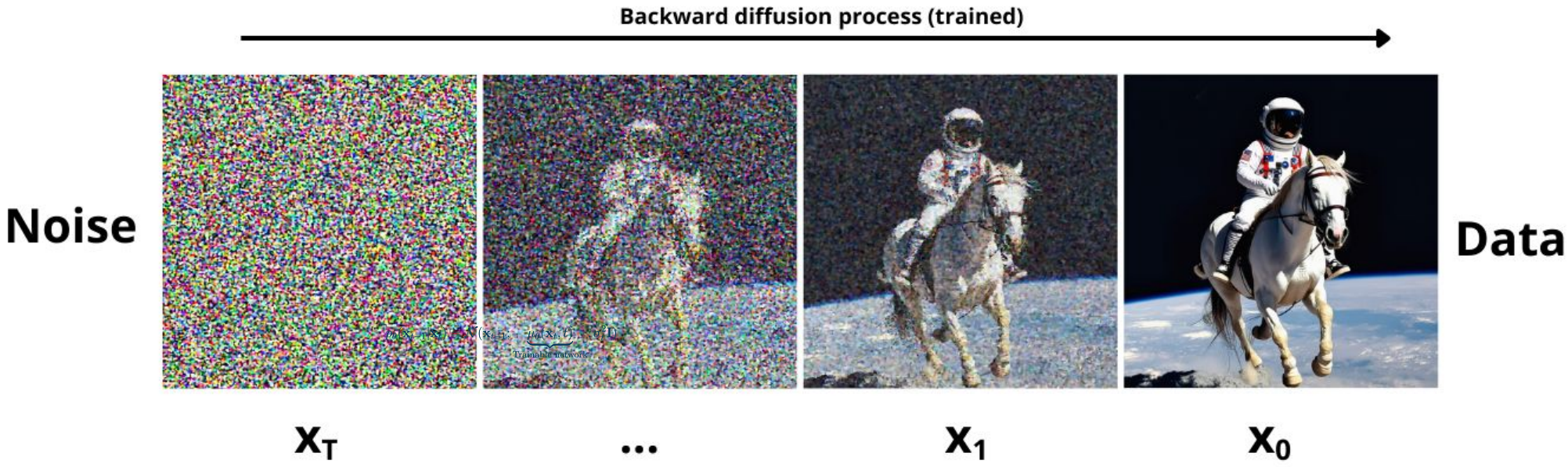


$$q(x_{t-1} \mid x_t) = \int q(x_{t-1} \mid x_t, x_0) q(x_0 \mid x_t)\, dx_0 \longrightarrow \text{intractable :(}$$

But can we approximate? **Yes! Use a normal distribution with a small step for adding noise.**

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Reverse (backward) process



$$p(\mathbf{x}_T) = \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$$

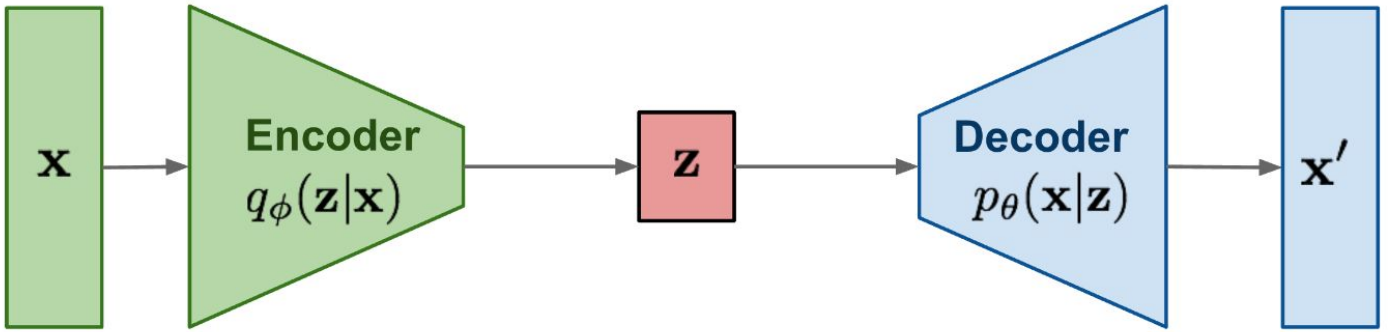
prior: pure noise

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\Big(\mu_\theta(x_t, t),\ \Sigma_\theta(x_t, t)\Big).$$

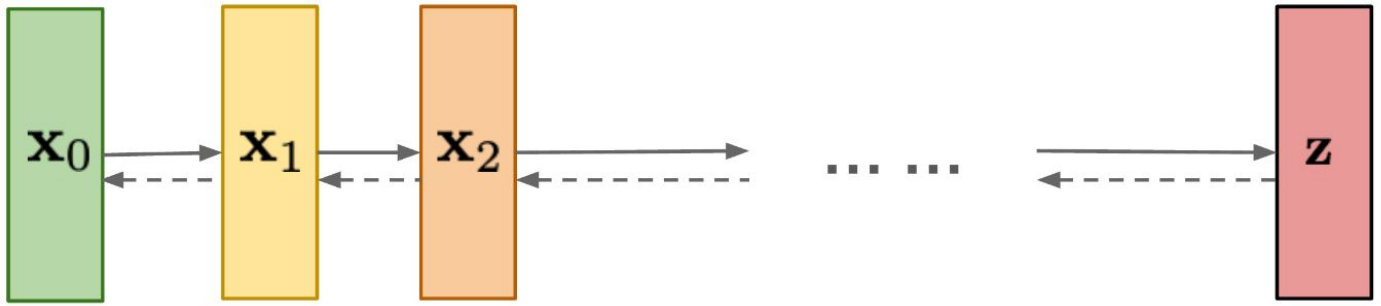
learned denoising step

Recap: VAE vs diffusion

VAE: maximize variational lower bound



Diffusion models: Gradually add Gaussian noise and then reverse



Diffusion models vs VAE

$$\max_{\theta} \mathbb{E}_{x_0 \sim q_{\text{data}}} \log p_\theta(x_0)$$

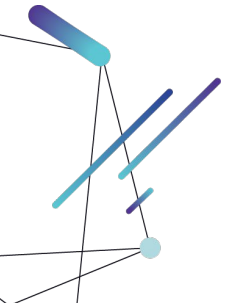
Diffusion models

$$ELBO_{DDPM} = \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}$$

VAE

$$ELBO_{VAE} = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x}_0)} \log \frac{p_\theta(\mathbf{x}_0, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x}_0)}$$

many latent states	LATENT STATES	one latent state
not trainable encoder	ENCODER	trainable encoder
same as data dim	LATENT DIM	differs from data dim

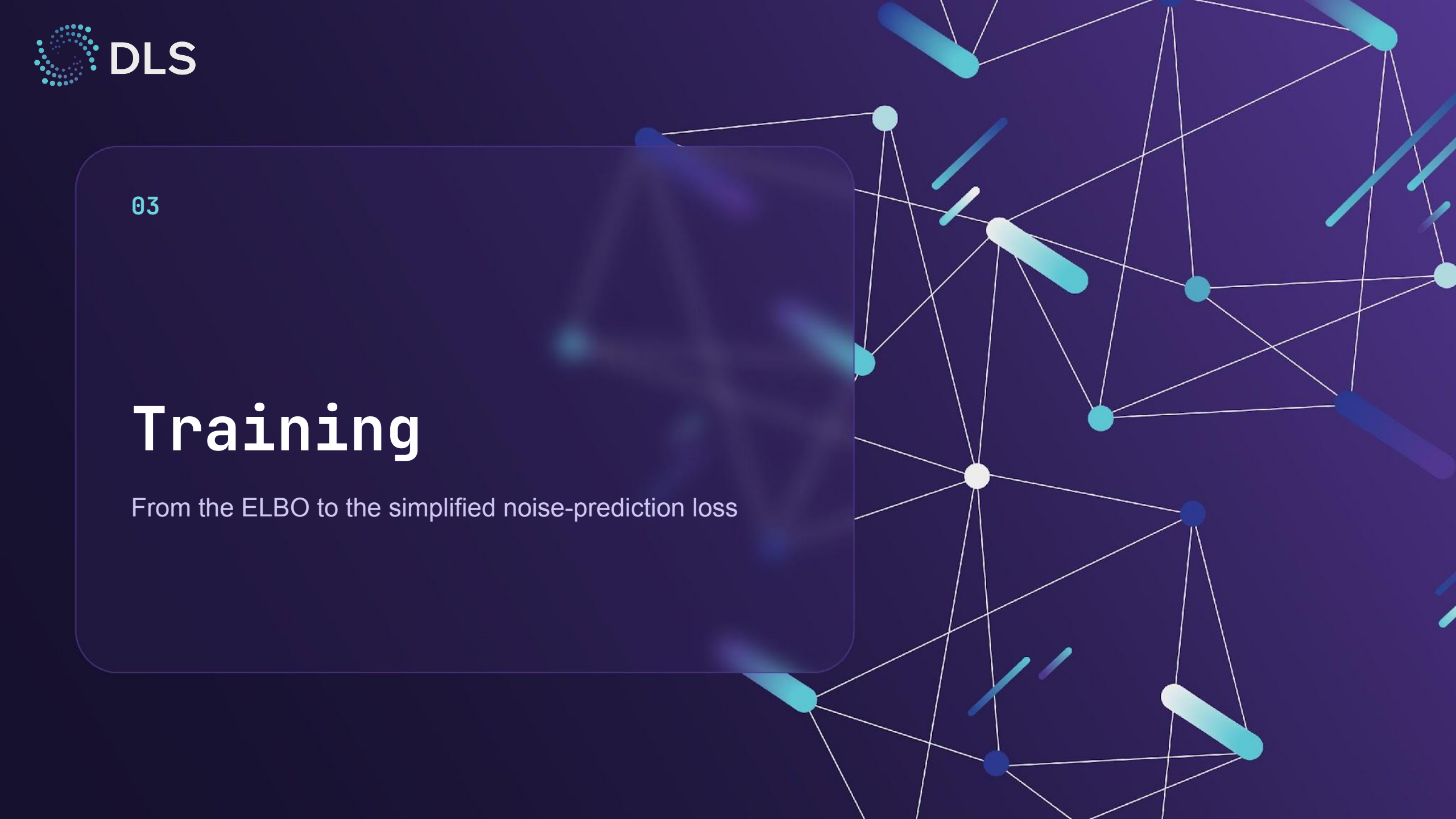


DLS

03

Training

From the ELBO to the simplified noise-prediction loss



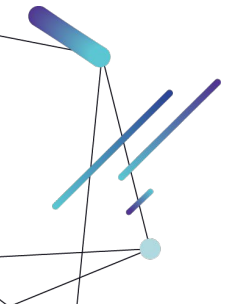
Training objective

ELBO loss

$$L = \mathbb{E}_q \Big[\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{L_T} + \sum_{t>1} \underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} \underbrace{- \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0} \Big]$$

FORWARD POSTERIOR — TRACTABLE

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\left(\tilde{\mu}_t(x_t, x_0),\ \tilde{\beta}_t I\right)$$



Training objective

ELBO loss

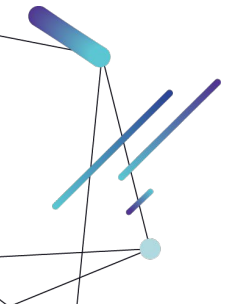
$$L = \mathbb{E}_q \Big[\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{L_T} + \sum_{t>1} \underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} \underbrace{- \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0} \Big]$$

FORWARD POSTERIOR — TRACTABLE

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\big(\tilde{\mu}_t(x_t, x_0),\ \tilde{\beta}_t I\big)$$

LEARNED REVERSE STEP

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\big(\mu_\theta(x_t, t),\ \Sigma_\theta(x_t, t)\big).$$



Forward posterior via Bayes' rule

If x_0 is known, the reverse step is a Gaussian with a known mean and variance

Bayes' rule

$$q(x_{t-1} \mid x_t, x_0) = \frac{q(x_t \mid x_{t-1})\, q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)}$$

ALL THREE TERMS ON THE RIGHT ARE KNOWN GAUSSIANS

forward step

$$q(x_t \mid x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}\, x_{t-1},\ \beta_t I)$$

marginal at t − 1

$$q(x_{t-1} \mid x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_{t-1}}\, x_0,\ (1-\bar{\alpha}_{t-1}) I)$$

marginal at t

$$q(x_t \mid x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\, x_0,\ (1-\bar{\alpha}_t) I)$$

RESULT

Product of Gaussians is Gaussian: mean and variance in closed form

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}(\tilde{\mu}_t(x_t, x_0),\ \tilde{\beta}_t I)$$

$$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}\,\beta_t}{1-\bar{\alpha}_t}\, x_0 + \frac{\sqrt{\alpha_t}\,(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\, x_t$$

$$\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\,\beta_t$$

Forward posterior via Bayes' rule

If x_0 is known, the reverse step is a Gaussian with a known mean and variance

Bayes' rule

$$q(x_{t-1} \mid x_t, x_0) = \frac{q(x_t \mid x_{t-1})\, q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)}$$

ALL THREE TERMS ON THE RIGHT ARE KNOWN GAUSSIANS

forward step

$$q(x_t \mid x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}\, x_{t-1},\ \beta_t I)$$

marginal at t − 1

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marginal at t

$$q(x_t \mid x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\, x_0,\ (1-\bar{\alpha}_t) I)$$

RESULT

Product of Gaussians is Gaussian: mean and variance in closed form

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}(\tilde{\mu}_t(x_t, x_0),\ \tilde{\beta}_t I)$$

$$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}\,\beta_t}{1-\bar{\alpha}_t}\, x_0 + \frac{\sqrt{\alpha_t}\,(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\, x_t$$

$$\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\,\beta_t$$

Training objective

ELBO loss

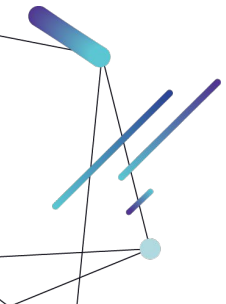
$$L = \mathbb{E}_q \Bigg[\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{L_T} + \sum_{t>1} \underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} \underbrace{- \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0} \Bigg]$$

FORWARD POSTERIOR — TRACTABLE

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\Big(\tilde{\mu}_t(x_t, x_0),\ \tilde{\beta}_t I\Big)$$

LEARNED REVERSE STEP

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\Big(\mu_\theta(x_t, t),\ \Sigma_\theta(x_t, t)\Big).$$



Training objective

ELBO loss

$$L = \mathbb{E}_q \Bigg[\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{L_T} + \sum_{t>1} \underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} \underbrace{- \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0} \Bigg]$$

FORWARD POSTERIOR — TRACTABLE

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\Big(\tilde{\mu}_t(x_t, x_0),\ \tilde{\beta}_t I\Big)$$

LEARNED REVERSE STEP

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\Big(\mu_\theta(x_t, t),\ \Sigma_\theta(x_t, t)\Big).$$

KL BETWEEN TWO GAUSSIANS

$$L_t = \mathbb{E}_{x_0,\epsilon}\left[\frac{1}{2\|\Sigma_\theta(x_t,t)\|_2^2}\|\tilde{\mu}_t(x_t,x_0) - \mu_\theta(x_t,t)\|^2\right]$$

Training objective

ELBO loss

$$L = \mathbb{E}_q \Big[\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{L_T} + \sum_{t>1} \underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} \underbrace{- \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0} \Big]$$

FORWARD POSTERIOR — TRACTABLE

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\big(\tilde{\mu}_t(x_t, x_0),\ \tilde{\beta}_t I\big)$$

LEARNED REVERSE STEP

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\big(\mu_\theta(x_t, t),\ \Sigma_\theta(x_t, t)\big).$$

FIX THE VARIANCE: ONLY THE MEAN IS LEARNED

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1};\ \underbrace{\mu_\theta(\mathbf{x}_t, t)}_{\text{Trainable network}},\ \sigma_t^2 \mathbf{I})$$

$$\Sigma_\theta(x_t, t) = \sigma_t^2 I$$

- the variance depends only on the timestep, not on the image
- it is the same for all coordinates

Training objective

ELBO loss

$$L = \mathbb{E}_q \Bigg[\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{L_T} + \sum_{t>1} \underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} \underbrace{- \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0} \Bigg]$$

FORWARD POSTERIOR — TRACTABLE

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\left(\tilde{\mu}_t(x_t, x_0),\ \tilde{\beta}_t I\right)$$

FIX THE VARIANCE: ONLY THE MEAN IS LEARNED

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1};\quad \underbrace{\mu_\theta(\mathbf{x}_t, t)}_{\text{Trainable network}}\quad , \sigma_t^2 \mathbf{I})$$

LOSS WITH FIXED VARIANCE

$$L_t = \mathbb{E}_{x_0,\epsilon}\left[\frac{1}{2\sigma_t^2}\left\|\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)\right\|^2\right]$$

DLS

From μ to ε: reparametrizing the mean

1 **Forward process in closed form**

express x_0 through x_t and the noise ε

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon \quad \Rightarrow \quad x_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(x_t - \sqrt{1-\bar{\alpha}_t}\, \epsilon\right)$$

2 **Posterior mean of $q(x_{t-1} \mid x_t, x_0)$**

tractable, depends on x_0 and x_t

$$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}\,\beta_t}{1-\bar{\alpha}_t}\, x_0 + \frac{\sqrt{\alpha_t}\,(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\, x_t$$

3 **Substitute x_0**

the mean is x_t minus scaled noise

$$\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\, \epsilon\right)$$

4 **Same form for the model**

the network predicts ε instead of μ

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\, \epsilon_\theta(x_t, t)\right)$$

RESULT

μ-matching becomes ε-matching (next slide)

$$\|\tilde{\mu}_t - \mu_\theta\|^2 = \frac{\beta_t^2}{\alpha_t\,(1-\bar{\alpha}_t)}\, \|\epsilon - \epsilon_\theta(x_t, t)\|^2, \qquad \beta_t = 1 - \alpha_t$$

Training objective

ELBO loss

$$L = \mathbb{E}_q \Bigg[\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{L_T} + \sum_{t>1} \underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} \underbrace{- \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0} \Bigg]$$

LOSS WITH FIXED VARIANCE

$$L_t = \mathbb{E}_{x_0,\epsilon} \left[\frac{1}{2\sigma_t^2} \left\| \tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t) \right\|^2 \right]$$

PREDICT THE NOISE E INSTEAD OF THE MEAN

$$L_t = \mathbb{E}_{x_0,\epsilon} \left[\frac{(1-\alpha_t)^2}{2\alpha_t(1-\bar{\alpha}_t)\sigma_t^2} \left\| \epsilon_t - \epsilon_\theta(x_t, t) \right\|^2 \right]$$

SIMPLIFIED OBJECTIVE

$$L_{simple,t} = \mathbb{E}_{x_0,\epsilon} \left[\left\| \epsilon_t - \epsilon_\theta(x_t, t) \right\|^2 \right]$$

Source

Summary: training & sampling

Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \text{Uniform}(\{1, \dots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on

$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$

6: **until** converged

Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \dots, 1$ **do**
3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

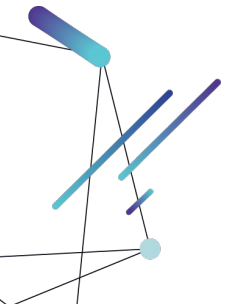
Source

Summary

DLS

+ Pros

- Good quality
- Diverse generation



Summary

+ Pros

- Good quality
- Diverse generation

- Cons

- Low speed — many denoising steps



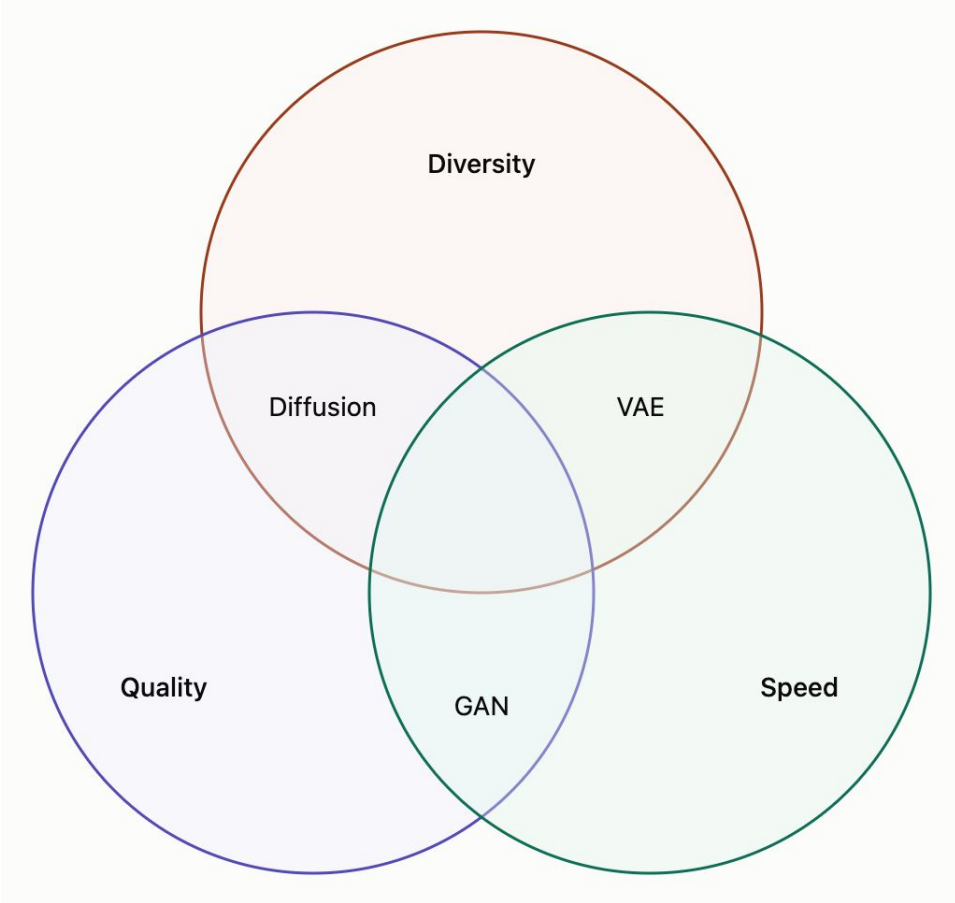
Summary: the generative trilemma

+ Pros

- Good quality
- Diverse generation

- Cons

- Low speed — many denoising steps

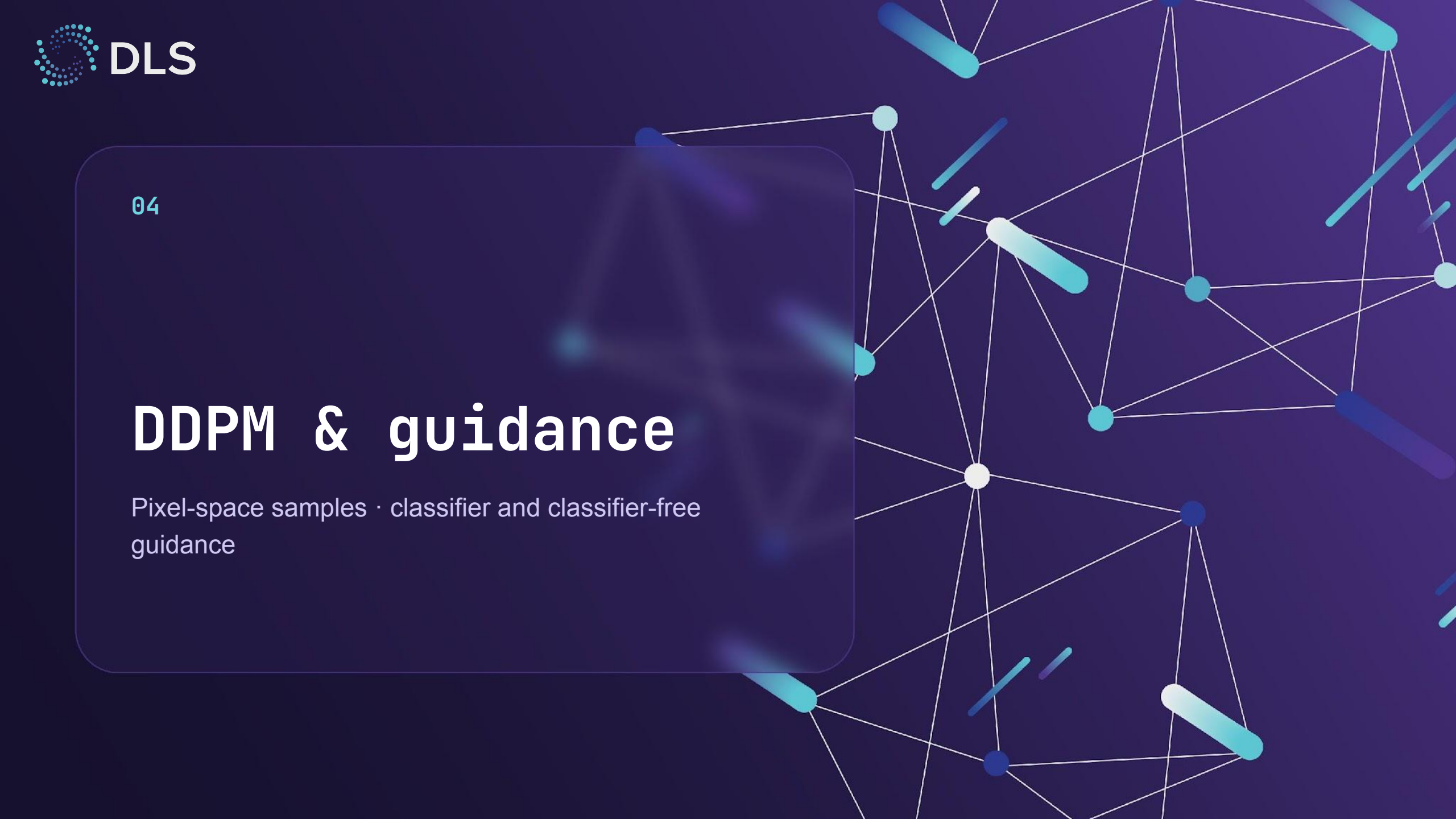


DLS

04

DDPM & guidance

Pixel-space samples · classifier and classifier-free guidance



DDPM: diffusion in pixel space

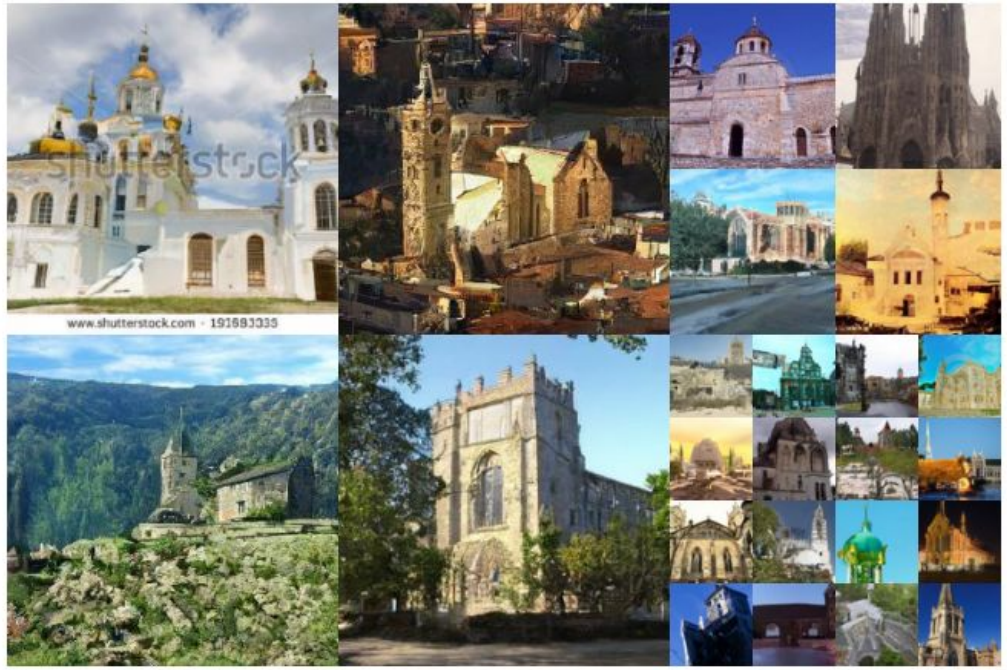


Figure 3: LSUN Church samples. FID=7.89

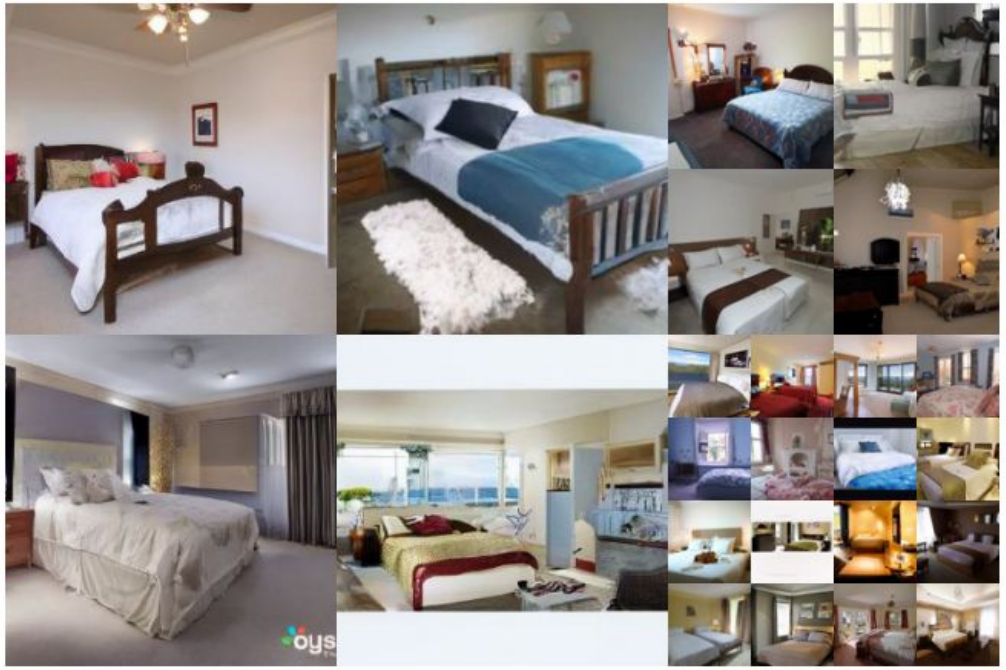


Figure 4: LSUN Bedroom samples. FID=4.90

Source

DDPM: diffusion in pixel space



Source

DDPM: diffusion in pixel space



LSUN 256×256

Church, Bedroom and Cat samples. Notice that the model occasionally generates dataset watermarks.

Source

Classifier guidance

Algorithm 1 Classifier guided diffusion sampling, given a diffusion model $(\mu_\theta(x_t), \Sigma_\theta(x_t))$, classifier $p_\phi(y|x_t)$, and gradient scale s.

Input: class label y, gradient scale s
$x_T \leftarrow$ sample from $\mathcal{N}(0, \mathbf{I})$
for all t from T to 1 **do**
  $\mu, \Sigma \leftarrow \mu_\theta(x_t), \Sigma_\theta(x_t)$
  $x_{t-1} \leftarrow$ sample from $\mathcal{N}(\mu + s\Sigma \nabla_{x_t} \log p_\phi(y|x_t), \Sigma)$
end for
return x_0

classifier gradient

1. Class-conditional modeling of $p(x_t|c)$
2. Train an extra classifier $p(c|x_t)$
3. Mix its gradient with the diffusion model during sampling

Source

Classifier-free guidance: training

Algorithm 1 Joint training a diffusion model with classifier-free guidance

Require: p_{uncond}: probability of unconditional training

1: **repeat**
2: $(\mathbf{x}, \mathbf{c}) \sim p(\mathbf{x}, \mathbf{c})$ ▷ Sample data with conditioning from the dataset
3: $\mathbf{c} \leftarrow \varnothing$ with probability p_{uncond} ▷ Randomly discard conditioning to train unconditionally
4: $\lambda \sim p(\lambda)$ ▷ Sample log SNR value
5: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
6: $\mathbf{z}_\lambda = \alpha_\lambda \mathbf{x} + \sigma_\lambda \boldsymbol{\epsilon}$ ▷ Corrupt data to the sampled log SNR value
7: Take gradient step on $\nabla_\theta \left\| \boldsymbol{\epsilon}_\theta(\mathbf{z}_\lambda, \mathbf{c}) - \boldsymbol{\epsilon} \right\|^2$ ▷ Optimization of denoising model
8: **until** converged

Source

Classifier-free guidance: sampling

Algorithm 2 Conditional sampling with classifier-free guidance

Require: w: guidance strength
Require: $\mathbf{c}$: conditioning information for conditional sampling
Require: $\lambda_1, \ldots, \lambda_T$: increasing log SNR sequence with $\lambda_1 = \lambda_{\min}, \lambda_T = \lambda_{\max}$

1: $\mathbf{z}_1 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = 1, \ldots, T$ **do**
  ▷ Form the classifier-free guided score at log SNR λ_t
3: $\tilde{\boldsymbol{\epsilon}}_t = (1 + w)\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, \mathbf{c}) - w\boldsymbol{\epsilon}_\theta(\mathbf{z}_t)$
  ▷ Sampling step (could be replaced by another sampler, e.g. DDIM)
4: $\tilde{\mathbf{x}}_t = (\mathbf{z}_t - \sigma_{\lambda_t}\tilde{\boldsymbol{\epsilon}}_t)/\alpha_{\lambda_t}$
5: $\mathbf{z}_{t+1} \sim \mathcal{N}(\tilde{\boldsymbol{\mu}}_{\lambda_{t+1}|\lambda_t}(\mathbf{z}_t, \tilde{\mathbf{x}}_t), (\tilde{\sigma}^2_{\lambda_{t+1}|\lambda_t})^{1-v}(\sigma^2_{\lambda_t|\lambda_{t+1}})^v)$ if $t < T$ else $\mathbf{z}_{t+1} = \tilde{\mathbf{x}}_t$
6: **end for**
7: **return** $\mathbf{z}_{T+1}$

Source

Classifier-free guidance: effect of scale w

no guidance



w = 1



w = 3



stronger guidance → higher fidelity, lower diversity

DLS

05

Text-to-image

GLIDE, Imagen, DALLE



GLIDE: text-guided generation



"a surrealist dream-like oil painting by salvador dalí of a cat playing checkers"



"a professional photo of a sunset behind the grand canyon"



"a high-quality oil painting of a psychedelic hamster dragon"



"an illustration of albert einstein wearing a superhero costume"



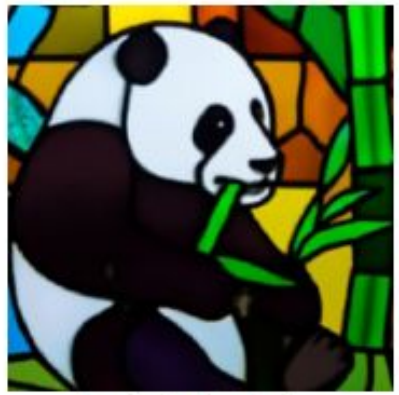
"a boat in the canals of venice"



"a painting of a fox in the style of starry night"



"a red cube on top of a blue cube"



"a stained glass window of a panda eating bamboo"

Source

GLIDE: CLIP guidance vs classifier-free

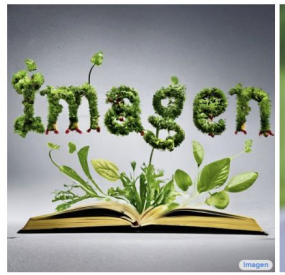


(c) GLIDE (CLIP guidance, scale 2.0)

(d) GLIDE (Classifier-free guidance, scale 3.0)

Source

Imagen: cascaded diffusion



Sprouts in the shape of text 'Imagen' coming out of a fairytale book.



A photo of a Shiba Inu dog with a backpack riding a bike. It is wearing sunglasses and a beach hat.



A high contrast portrait of a very happy fuzzy panda dressed as a chef in a high end kitchen making dough. There is a painting of flowers on the wall behind him.



Teddy bears swimming at the Olympics 400m Butterfly event.



A cute corgi lives in a house made out of sushi.



A cute sloth holding a small treasure chest. A bright golden glow is coming from the chest.



A brain riding a rocketship heading towards the moon.



A dragon fruit wearing karate belt in the snow.



A strawberry mug filled with white sesame seeds. The mug is floating in a dark chocolate sea.



A relaxed garlic with a blindfold reading a newspaper while floating in a pool of tomato soup.



A photo of a corgi dog wearing a wizard hat playing guitar on the top of a mountain.



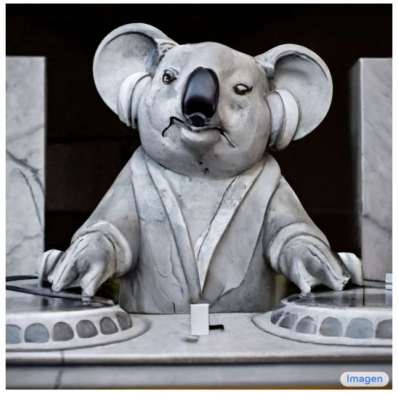
A single beam of light enter the room from the ceiling. The beam of light is illuminating an easel. On the easel there is a Rembrandt painting of a raccoon.



A squirrel is inside a giant bright shiny crystal ball in on the surface of blue ocean. There are few clouds in the sky.



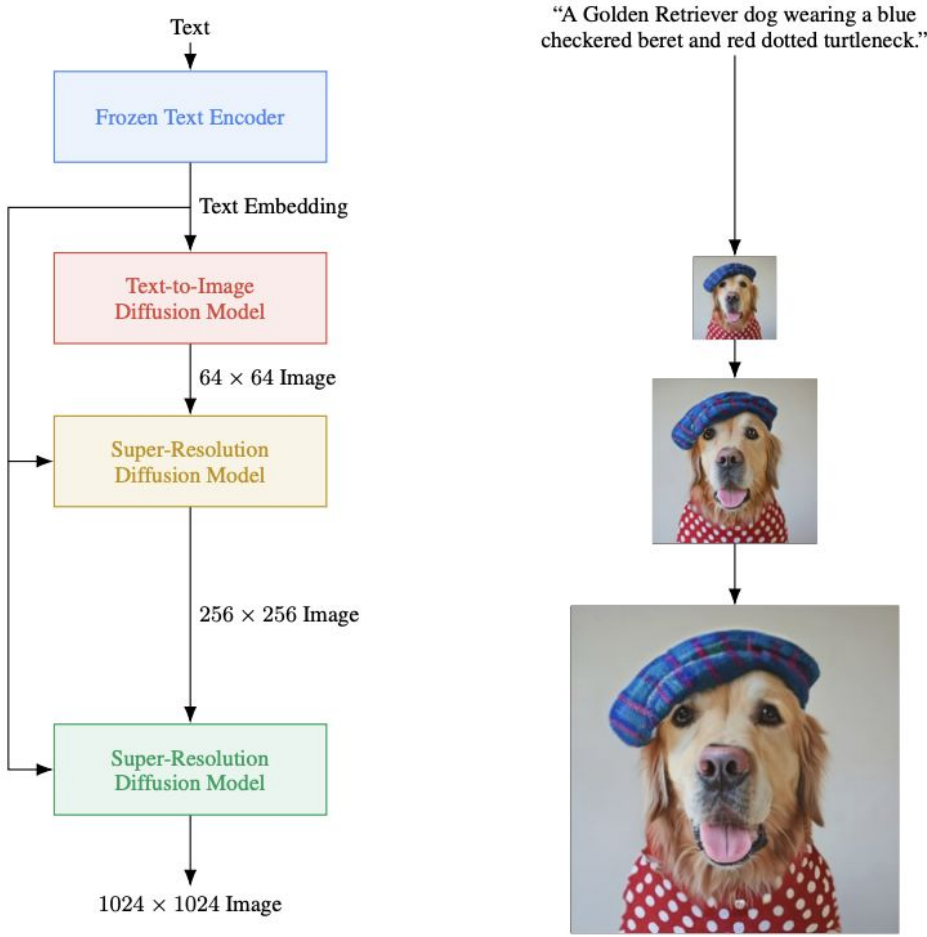
A bald eagle made of chocolate powder, mango, and whipped cream.



A marble statue of a Koala DJ in front of a marble statue of a turntable. The Koala has wearing large marble headphones.

Source

Imagen: cascaded diffusion



1 Frozen text encoder

T5-XXL turns the prompt into text embeddings

2 Base diffusion model

generates a 64 × 64 image from the text

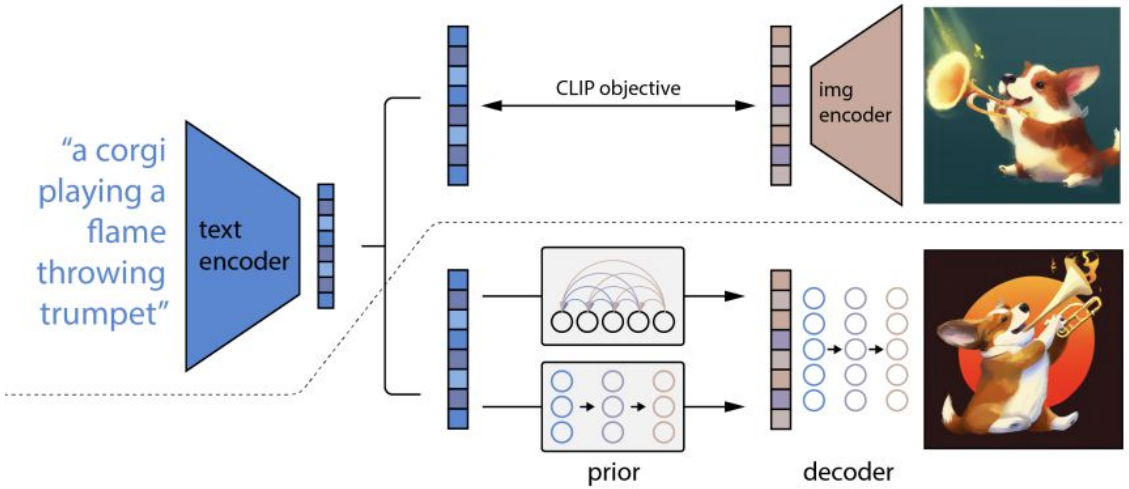
3 Super-resolution ×2

two diffusion models upsample to 256 and then 1024 px

Source

paper →

DALL·E 2 (unCLIP)

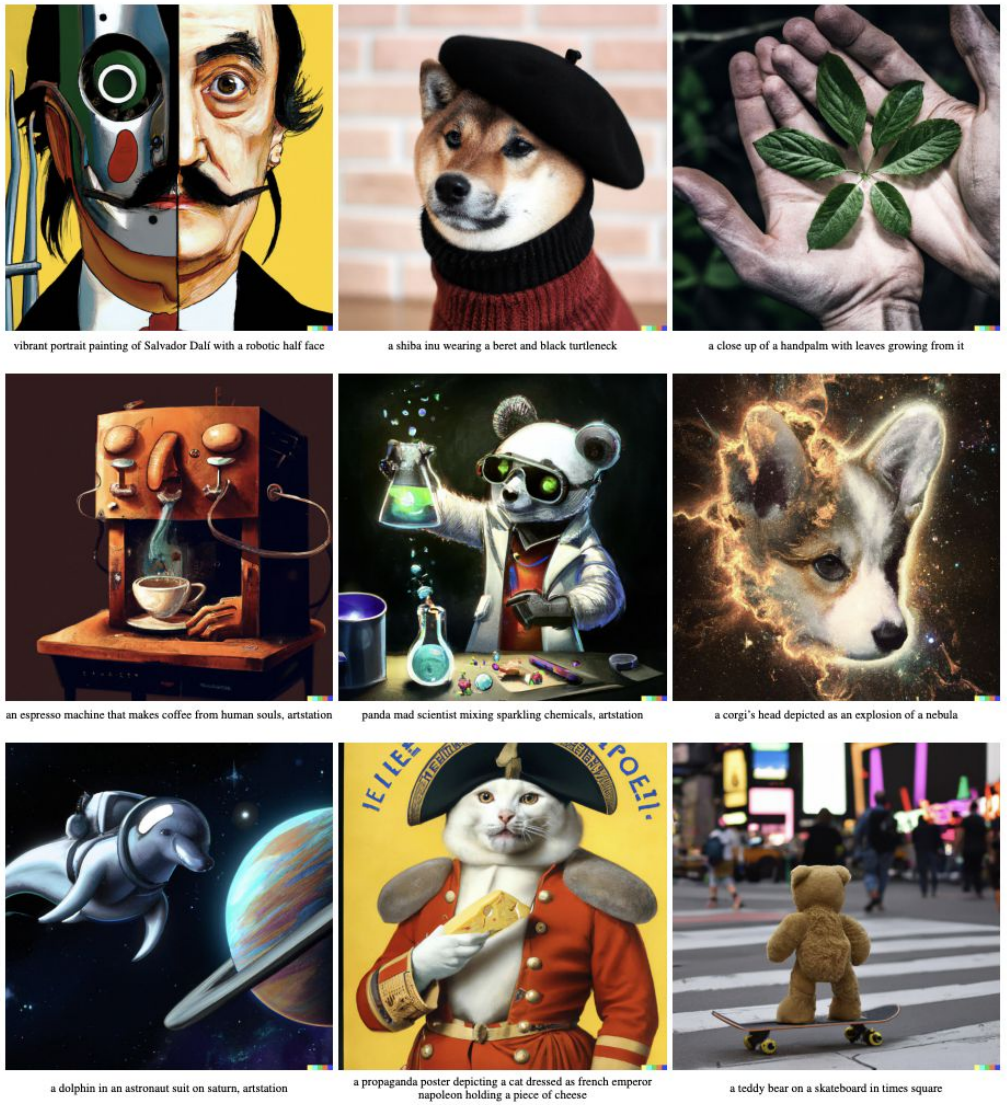


Why a prior?

- Generating the image embedding improves diversity with minimal loss in photorealism
- Guidance only on the decoder

Editing in CLIP space

- Variations of an input image
- Interpolations
- Text diffs



DLS



Коновалова Нина

@ReadingAI

@AfeliaN

Thank you for attention!

@READING_AI